



DIFFERENT DATABASES WITH EXAMPLES

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DATA TYPES

Structured Data

Definition:

Organized data that follows a predefined format and schema, stored in rows and columns.

Characteristics:

- Fixed format with clear relationships
- Easily searchable and queryable
- Follows strict rules and constraints
- Highly organized and predictable

Example:

CustomerID	Name	Email	Phone
1	John Smith	john@email.com	555-0123
2	Sarah Johnson	sarah@email.com	555-0456
3	Mike Brown	mike@email.com	555-0789

Unstructured Data

Definition:

Data without a predefined format or organization, existing in its natural form.

Characteristics:

- No fixed schema or format
- Flexible and varied content
- Requires processing for analysis
- Rich in context and meaning

Examples:

Social Media Post:

```
{
  "user": "alex_photographer",
  "timestamp": "2024-06-28T14:30:00Z",
  "content": "Just captured this amazing sunset! #photography #nature",
  "image": "sunset_photo.jpg",
  "location": "Santa Monica Beach",
  "likes": 127,
  "comments": ["Beautiful shot!", "Love the colors!", "Where is this?"]
}
```

DATA TYPES

Email Message:

From: marketing@company.com
To: newsletter@subscribers.com
Subject: Summer Sale - 50% Off Everything!
Date: June 28, 2024, 2:15 PM

Dear Valued Customer,

We're excited to announce our biggest sale of the year!
[Attached: summer_catalog.pdf, promo_banner.jpg]

Unstructured Text Document:

Meeting Notes - June 28, 2024
Project Alpha Discussion

- Budget approval pending (waiting for CFO review)
- Timeline: Launch target moved to Q3
- Team feedback: Need 2 additional developers
- Next steps: Schedule follow-up with stakeholders
- Action items assigned to various team members

DATABASES FOR STRUCTURED DATA

1. RELATIONAL DATABASES (SQL)

What it is:

Tables with rows, columns, and relationships

When to use: Need ACID compliance, complex queries, data consistency

Example use case: Banking transactions, e-commerce orders

Data Example:

CustomerID	Name	Email	Phone
1	John Smith	john@email.com	555-0123
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DATABASES FOR STRUCTURED DATA

MySQL

- Web applications, e-commerce
- Example: Online store product catalog

PostgreSQL

- Enterprise apps, geospatial data
- Example: Ride-sharing location tracking

Oracle

- Large enterprise, mission-critical systems
- Example: Airline reservation systems

SQL Server

- Microsoft ecosystem, enterprise applications
- Example: Corporate ERP systems

2. NEWSQL DATABASES

What it is:

ACID properties with horizontal scaling

When to use: Global applications needing both consistency and scale

Example use case: Multi-region financial applications

Data Example:

GLOBAL_TRANSACTIONS TABLE (distributed across regions):					
ID	From	To	Amount	Currency	Region
1	ACC001	ACC002	1000	USD	US-East
2	ACC003	ACC004	500	EUR	EU-West

CockroachDB

- Distributed SQL with ACID guarantees
- Example: Global banking applications

TiDB

- MySQL-compatible distributed database
- Example: Large-scale OLTP applications

DATABASES FOR STRUCTURED DATA

3. TIME-SERIES DATABASES

What it is:

Optimized for time-stamped structured data

When to use: Monitoring, IoT sensors, financial data

Example use case: Server metrics, stock prices

Data Example:

SERVER_METRICS:

timestamp	cpu_usage	memory_us	disk_io
2024-01-15 10:00:00	45.20%	78.50%	120MB/s
2024-01-15 10:01:00	52.80%	79.10%	135MB/s
2024-01-15 10:02:00	41.30%	77.90%	98MB/s

InfluxDB

- Purpose-built for time-series
- Example: Tesla vehicle telemetry data

TimescaleDB

- PostgreSQL extension for time-series
- Example: Financial trading data analysis

Examples of structured data



Pricing data



CRM data



Dates & times



Financial transactions



Customer account data



Medical information

DATABASES FOR UNSTRUCTURED DATA

1. DOCUMENT DATABASES



What it is:

Stores data as documents (JSON-like)

When to use: Flexible schema, rapid development, varied data structures

Example use case: Content management, user profiles

Data Example:

```
{
  "_id": "user123",
  "name": "John Doe",
  "age": 30,
  "preferences": {
    "themes": ["sci-fi", "comedy"],
    "languages": ["English", "Spanish"]
  },
  "viewing_history": [
    {
      "movie": "Inception",
      "watched_date": "2024-01-15",
      "rating": 4.5
    },
    {
      "series": "Stranger Things",
      "season": 4,
      "episodes_watched": [1, 2, 3]
    }
  ],
  "subscription": {
    "type": "premium",
    "expires": "2024-12-31"
  }
}
```

MongoDB

- Most popular document database
- Example: Netflix user preferences and viewing history

CouchDB

- Multi-master replication, offline-first
- Example: Mobile apps with sync capabilities

DATABASES FOR UNSTRUCTURED DATA

2. KEY-VALUE DATABASES



What it is:

Simple key-value pairs for storage

When to use: Caching, session storage, high-speed lookups

Example use case: Gaming leaderboards, user sessions

Data Example:

Key	Value
user:session:abc123	{"user_id": 456, "login_time": "2024-01-15T10:30:00", "last_activity": "2024-01-15T11:45:00"}
leaderboard:top10	[{"player": "ProGamer", "score": 98500}, {"player": "SkillMaster", "score": 97200}]
cache:product:123	{"name": "Gaming Mouse", "price": 79.99, "stock": 15}
config:app:settings	{"theme": "dark", "notifications": true, "language": "en"}

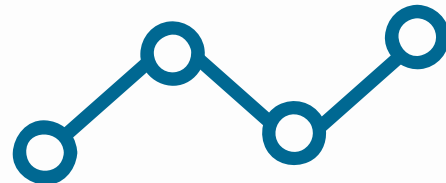
Redis

- In-memory, extremely fast
- Example: Real-time chat applications

DynamoDB

- AWS managed, serverless scaling
- Example: IoT device data collection

3. GRAPH DATABASES



What it is:

Nodes and relationships for connected data

When to use: Social networks, fraud detection, recommendations

Example use case: Friend connections, supply chain tracking

Data Example:

Nodes:

```
(Alice:Person {name: "Alice", age: 28, city: "NYC"})
(Bob:Person {name: "Bob", age: 32, city: "LA"})
(TechCorp:Company {name: "TechCorp", industry: "Software"})
(DataSci:Skill {name: "Data Science", category: "Technical"})
```

Relationships:

```
(Alice)-[:FRIENDS_WITH {since: "2020-01-15"}]->(Bob)
(Alice)-[:WORKS_AT {position: "Data Analyst", since: "2023-03-01"}]->(TechCorp)
(Bob)-[:WORKS_AT {position: "Developer", since: "2022-01-15"}]->(TechCorp)
(Alice)-[:HAS_SKILL {level: "Advanced"}]->(DataSci)
(Bob)-[:KNOWS {through: "work"}]->(Alice)
```

DATABASES FOR UNSTRUCTURED DATA

Neo4j

- Most popular graph database
- Example: LinkedIn professional networks

Amazon Neptune

- Managed graph service
- Example: Fraud detection in financial transactions

4. VECTOR DATABASES

What it is:

Stores high-dimensional vectors for similarity search

When to use: AI/ML applications, semantic search, recommendations

Example use case: ChatGPT context retrieval, image similarity

Data Example:

```
{
  "id": "doc_001",
  "text": "Machine learning is a subset of artificial intelligence",
  "vector": [0.1, -0.2, 0.8, 0.3, -0.1, 0.9, ...], // 1536 dimensions
  "metadata": {
    "source": "AI textbook",
    "chapter": "Introduction to ML",
    "topic": "machine_learning"
  }
}

{
  "id": "doc_002",
  "text": "Deep learning uses neural networks with multiple layers",
  "vector": [0.2, -0.1, 0.7, 0.4, -0.2, 0.8, ...], // 1536 dimensions
  "metadata": {
    "source": "AI textbook",
    "chapter": "Neural Networks",
    "topic": "deep_learning"
  }
}
```

FAISS

- High-performance similarity search library by Facebook AI
- Example: Fast nearest neighbor search for large-scale image retrieval

Qdrant

- Open-source vector database optimized for production
- Example: Real-time personalized recommendations in e-commerce

Pinecone

- Managed vector database
- Example: Product recommendations based on user behavior

DATABASES FOR UNSTRUCTURED DATA

Chroma

- Open-source, embeddings focused
- Example: Document similarity search

Weaviate

- Vector database with ML models
- Example: Semantic search in knowledge bases

5. SEARCH DATABASES

What it is:

Full-text search and analytics engine

When to use: Search functionality, log analysis, text analytics

Example use case: E-commerce search, application logs

Data Example:

```
{
  "_index": "products",
  "_id": "prod123",
  "_source": {
    "title": "Wireless Bluetooth Headphones",
    "description": "Premium noise-canceling headphones with 30-hour battery life",
    "category": "Electronics",
    "brand": "AudioTech",
    "price": 199.99,
    "tags": ["wireless", "bluetooth", "noise-canceling", "premium"],
    "reviews": [
      {
        "rating": 5,
        "comment": "Amazing sound quality and comfort"
      },
      {
        "rating": 4,
        "comment": "Great battery life, worth the price"
      }
    ],
    "searchable_text": "wireless bluetooth headphones premium noise canceling audiotech amazing sound quality comfort battery life"
  }
}
```

Elasticsearch

- Distributed search and analytics
- Example: Amazon product search

Solr

- Apache Lucene-based search
- Example: Enterprise document search

DATABASES FOR UNSTRUCTURED DATA

6.COLUMN-FAMILY DATABASES

What it is:

Data stored in column families, not rows

When to use: High write volumes, big data analytics

Example use case: Log analysis, social media feeds

Data Example:

Column Family: user_activity
Row Key: user123_2024-01-15

Standard Columns:
login_time: "10:30:00"
logout_time: "14:45:00"
device: "mobile"

Dynamic Columns (time-based):
activity:10:31:00: "viewed_homepage"
activity:10:32:15: "searched_products"
activity:10:35:22: "added_to_cart"
activity:11:10:05: "completed_purchase"
activity:14:30:18: "viewed_profile"

Cassandra

- Distributed, high availability
- Example: Spotify music streaming logs

HBase

- Hadoop ecosystem, big data processing
- Example: Social media activity feeds

DATABASES FOR BOTH DATA TYPES

MULTI-MODEL DATABASES

What it is:

Supports multiple data models in one system

When to use: Applications with diverse data requirements

Example use case: E-commerce with products, users, and recommendations

Data Example:

```
// Document model for user profiles
{
  "_id": "user456",
  "name": "Sarah Johnson",
  "preferences": {"color": "blue", "size": "M"}
}
```

```
// Graph model for recommendations
(Sarah)-[:BOUGHT]->(ProductA)
(Sarah)-[:VIEWED]->(ProductB)
(ProductA)-[:SIMILAR_TO]->(ProductC)
```

```
// Key-value for session data
session:user456 = {"cart": ["item1", "item2"], "last_login": "2024-01-15"}
```

ArangoDB

- Document, graph, and key-value in one
- Example: Social commerce platform

Azure Cosmos DB

- Microsoft's globally distributed multi-model
- Example: Gaming applications with multiple data types